1-8	9	10	11	12	13	14	15
15	10	10	9	2	4	0	0

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Martinho Santos Marques UPF ID: U 2 7 3 9 2 9

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   $\frac{3 - \mu_0}{1} = 1.96$

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   $SE(\bar{Y}) = \frac{\sigma_Y}{\sqrt{n}}$

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
17'5 | 10 | 8 | 1 | 12 | 7 | 5 | 2

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Alba Muñoz UPF ID: U273925

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

$E[Y_1] = \bar{Y}$ = unbiased

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   $\frac{3-\mu}{1} \leq 1.96$

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

   $[1.04, 4.96]$

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 10 | 10 | 12 | 1 | 6 | 5 | 0

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Xavier Esteban Lario UPF ID: U 2 7 3 9 1 0

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$. ← THIS ONE
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   $\sqrt{100} = 10$
   $\sqrt{400} = 20$

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 10 | 10 | 12 | 1 | 6 | 0 | 5

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Jordi Oliver UPF ID: U273944

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

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   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
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   (c) $U_i = \widehat{U}_i$ for every observation.
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   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

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   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
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   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.